CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Tue, Oct 13, 2026, 09:17 PM
Patient ID: CV-HIGHRISK-99
Architecture: 57-Feature Four-Model Architecture
Active Mode: Standard Production Mode

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Arthur Pendelton (High-Risk Profile)	**CV-HIGHRISK-99**	**75 yrs · Male**	**172 cm / 92 kg (31.1 kg/m²)**

1. Cardiovascular Prediction Summary

Multi-vessel predictive classification derived from the 57-dimensional clinical feature vector.

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
100.0%	**84.0%**	**58.0%**	**99.8%**
CAD	**Stenotic**	**Stenotic**	**Stenotic**
cad_model.pkl Threshold: **50.0%**	lad_model.pkl Threshold: **50.0%**	lcx_model.pkl Threshold: **50.0%**	rca_model.pkl Threshold: **50.0%**
High Est. Prob (≥70%)	**High Est. Prob (≥70%)**	**Moderate Est. Prob (50–69%)**	**High Est. Prob (≥70%)**

Model Estimation Notice: Values are model-estimated statistical classification probabilities, not clinically calibrated epidemiological risk scores or measured anatomical luminal stenosis narrowing percentages.

2. Clinical Input Measurements Summary

Patient clinical parameters from active evaluation session evaluated against informational reference benchmarks.

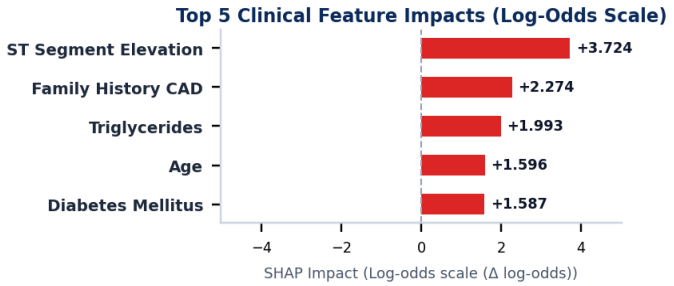
Clinical Parameter	Patient Value	General Reference Benchmark	Informational Evaluation Status
Resting Systolic BP (BP)	**165 mmHg**	< 120 mmHg (resting systolic)	Above reference (≥ 130 mmHg, Informational)
Resting Pulse Rate (PR)	**95 bpm**	60 – 100 bpm	Within standard resting range
Left Ventricular EF (EF-TTE)	**35%**	≥ 50%	LVEF < 50% (Below standard reference, Informational)
Fasting Blood Sugar (FBS)	**180 mg/dL**	70 – 99 mg/dL	Above threshold (≥ 126 mg/dL, Informational)
Serum Creatinine (CR)	**2.1 mg/dL**	0.7 – 1.3 mg/dL	Outside displayed adult reference range (Informational)
LDL Cholesterol	**190 mg/dL**	< 100 mg/dL	Above standard reference (≥ 100 mg/dL, Informational)
HDL Cholesterol	**28 mg/dL**	> 40 mg/dL (M) / > 50 mg/dL (F)	Below reference (< 40 mg/dL, Informational)
Triglycerides (TG)	**320 mg/dL**	< 150 mg/dL	Above reference (≥ 150 mg/dL, Informational)
Evaluated Clinical Vector	**57 Model Inputs**	Standardized 7-domain vector	34 numeric + 23 binary clinical indicators synchronized

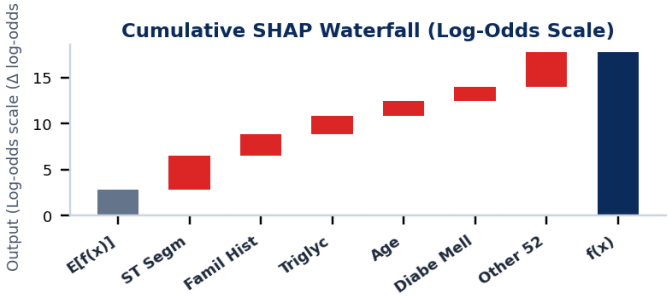
3. Patient-Specific SHAP Explainability (CAD & LAD)

Feature contributions calculated via game-theoretic Shapley values. **Scale Note:** TreeExplainer outputs additive probability contributions (Δ probability). LinearExplainer outputs additive log-odds contributions (Δ log-odds). Scales are strictly separated.

CAD Model Explainability (cad_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **CAD** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **2.810 log-odds** | Final Patient Output f(x): **17.717 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

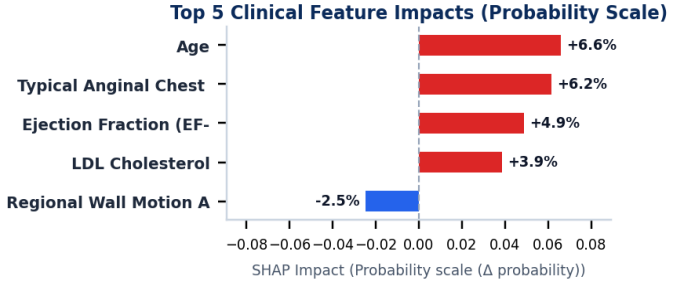
Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	ST Segment Elevation	ECG & Telemetry	Present (1)	+3.724 log-odds	Increases Risk
#2	Family History CAD	Medical History	Present (1)	+2.274 log-odds	Increases Risk
#3	Triglycerides	Biomarkers	320 mg/dL	+1.993 log-odds	Increases Risk
#4	Age	Basic Demographics	75 years	+1.596 log-odds	Increases Risk
#5	Diabetes Mellitus	Medical History	Present (1)	+1.587 log-odds	Increases Risk

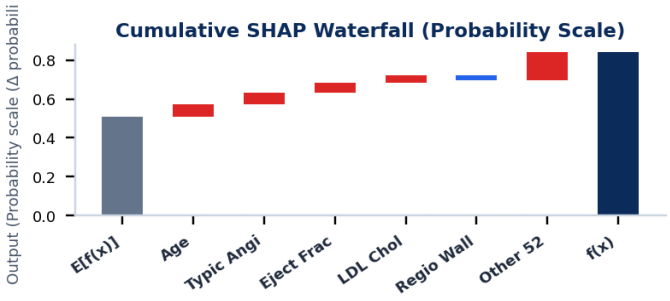




LAD Artery Model Explainability (lad_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.7%** | Final Patient Output f(x): **84.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Age	Basic Demographics	75 years	+6.6%	Increases Risk
#2	Typical Anginal Chest Pain	Symptoms	Present (1)	+6.2%	Increases Risk
#3	Ejection Fraction (EF-TTE)	Vital Signs	35 %	+4.9%	Increases Risk
#4	LDL Cholesterol	Biomarkers	190 mg/dL	+3.9%	Increases Risk
#5	Regional Wall Motion Abnormality	Echocardiography	Absent (0)	-2.5%	Decreases Risk



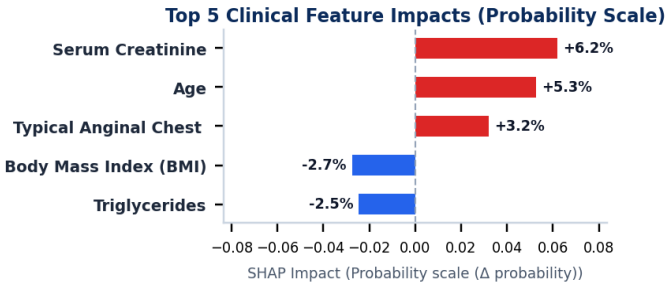


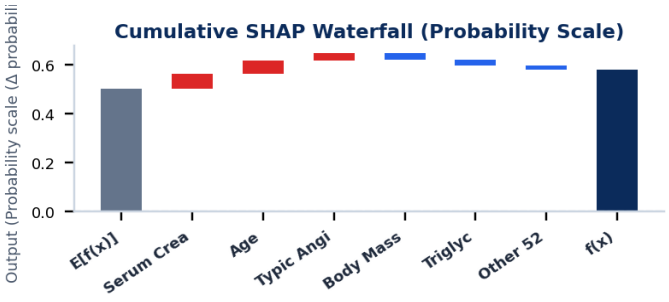
3. Patient-Specific SHAP Explainability (LCX & RCA)

Candidate and baseline branch vessel models with explicit scale separation and additivity verification.

LCX Artery Model Explainability (lcx_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.2%** | Final Patient Output f(x): **58.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

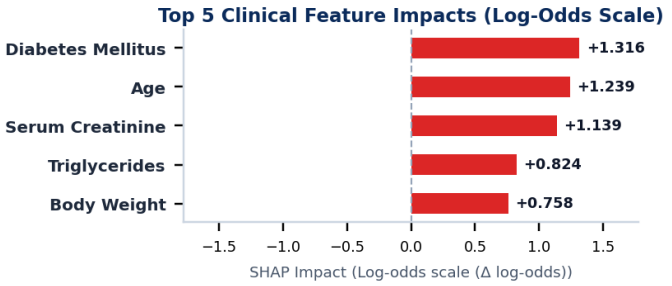
Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Serum Creatinine	Biomarkers	2.1 mg/dL	+6.2%	Increases Risk
#2	Age	Basic Demographics	75 years	+5.3%	Increases Risk
#3	Typical Anginal Chest Pain	Symptoms	Present (1)	+3.2%	Increases Risk
#4	Body Mass Index (BMI)	Basic Demographics	31.1 kg/m²	-2.7%	Decreases Risk
#5	Triglycerides	Biomarkers	320 mg/dL	-2.5%	Decreases Risk

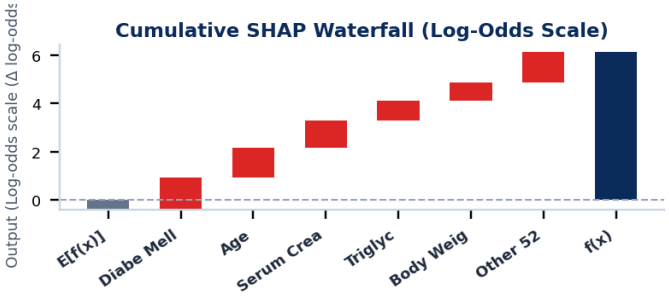




RCA Artery Model Explainability (rca_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **Stenotic** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **-0.393 log-odds** | Final Patient Output f(x): **6.133 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	Diabetes Mellitus	Medical History	Present (1)	+1.316 log-odds	Increases Risk
#2	Age	Basic Demographics	75 years	+1.239 log-odds	Increases Risk
#3	Serum Creatinine	Biomarkers	2.1 mg/dL	+1.139 log-odds	Increases Risk
#4	Triglycerides	Biomarkers	320 mg/dL	+0.824 log-odds	Increases Risk
#5	Body Weight	Basic Demographics	92 kg	+0.758 log-odds	Increases Risk





4. 3D Coronary Heart Visualization

High-resolution anatomical perspective from the interactive Three.js 106-mesh BodyParts3D cardiac model.



5. Model Performance and Reliability

Standard Mode: Retrospective model evaluation metrics recorded during development on held-out test data.

Model	Active Model File	Target Vascular Outcome	Accuracy	F1 Score	ROC-AUC
CAD Model	cad_model.pkl	Overall Coronary Artery Disease	**85.2%**	89.7%	**0.917**
LAD Model	lad_model.pkl	Left Anterior Descending Stenosis	**65.6%**	67.7%	**0.756**
LCX Model	lcx_model.pkl	Left Circumflex Stenosis	**57.4%**	53.6%	0.584
RCA Model	rca_model.pkl	Right Coronary Artery Stenosis	**60.7%**	50.0%	0.652

Model Reliability Notice: Standard LCX and RCA baseline models exhibit limited discriminatory power in development cohort testing (ROC-AUC 0.584 and 0.652). Probabilities must be interpreted with extreme caution and never guide clinical interventions independently.

6. Research Limitations, Educational Guidance & Disclaimer

Scope of Research Evaluation & Clinical Considerations:
- Model estimates represent statistical pattern matching against retrospective cohorts, not definitive clinical diagnoses.
- Results must always be interpreted in full clinical context by a licensed medical practitioner.
- Machine learning probabilities cannot substitute for direct coronary angiography, CT angiography, or stress imaging.
- Diagnostic and management decisions depend on patient presentation, physical exam, and comprehensive clinical history.

RESEARCH PROTOTYPE — NOT A CERTIFIED MEDICAL DEVICE:
This software and report are intended strictly for educational, scientific research, and methodology evaluation purposes. They are not approved or cleared by any regulatory health agency (e.g., FDA, CE, PMDA) for clinical diagnostic or therapeutic decision-making. Predictions and SHAP explainability analyses do not substitute for formal cardiovascular imaging, laboratory testing, or physician consultation.